

Hey, everybody. Welcome to your Chapter 20 review. Today, we will cover resistance training, our final component of integrated training. Creating and implementing resistance training programs for clients may be more involved and complicated than initially realized. Careful consideration must be taken to adequately meet the many facets of exercise principles, including any manipulation of training variables throughout the planned program.

You need to ensure that your workouts produce the desired outcomes for each individual. Programs that fail are at risk for decreased exercise adherence and increased risk of injury. Resistance training concepts are actually very scientific and detailed. It's critical to understand and be able to explain the principles of adaptation, progressive overload, and specificity, characteristics of strength development, and the various types of training systems, implementation, and exercise progressions. There's a lot more to it than shows up on the surface, right?

Your clients might think they know more than they do about resistance training, so educating them will be critical along the way. It's a popular form of training that people get into for a variety of reasons, and they may have developed preconceived notions about it. Knowing the content inside and out will allow you to educate your clients and guide them through safe and effective resistance training programs.

Ultimately, what resistance training really does is challenge and stress the neuromuscular system, acting as a catalyst for developing or maintaining lean body mass or fat-free mass. Whether your clients are looking to change body composition, enhance strength, athletic performance, or durability, resistance training can help.

Once you're done with this chapter, you should be able to summarize how resistance training affects the human body; define resistance training systems; identify the acute variables of training; categorize resistance exercises for different client types, goals, and adaptations; and employ methods to safely execute, instruct, and cue resistance exercises. To start, let's talk about how resistance training works. It's all about adaptation.

To achieve optimal training results, the body has to adapt to specific demands and stresses imposed on it. The ability to adapt to stress is known as the General Adaptation Syndrome, or GAS, as we call it. GAS includes three stages-- alarm reaction, resistance development, and exhaustion.

The alarm reaction stage is the initial reaction to a stressor, typically occurring 6 to 48 hours after initial stimulus to the neuromuscular system. This stage can include fatigue, joint stiffness, or delayed onset muscle soreness. The resistance development stage involves numerous physiological changes that ultimately lead to training adaptations that promote increases in performance. That's really the sweet spot of resistance training.

After repeated training sessions, the body increases its capability to efficiently recruit muscle fibers and distribute oxygen and blood to proper areas in the body. But what happens when you push past that point? Well, prolonged amounts of stress without adequate recovery can lead to the exhaustion stage, which may result in a breakdown within different systems of the body and subsequently contribute to things like stress fractures, muscle strains, ligament sprains, joint pain, and emotional fatigue. Incorporating sufficient repair and regeneration time, systematic progressions, and exercise variability into programs will help your clients avoid this stage.

Different tissues in the body will respond to different stresses. We see this in the principle of specificity, or SAID, which stands for Specific Adaptation to Imposed Demands. This principle describes the body's responses and adaptations to exercise and training. For example, someone repeatedly lifting heavy weights will produce higher levels of maximal strength, whereas a person repeatedly lifting lighter weights for a greater number of repetitions will develop higher levels of muscular endurance.

There are different types of specificity, with three in particular you should be aware of and are highly relevant to training. Mechanical specificity refers to the weight and movements placed on the body. Neuromuscular specificity refers to the speed of contraction and exercise selection. And metabolic specificity refers to the energy demands placed on the body. You need to ensure your programs provide a variety of intensities and stresses. That way, you can optimize the adaptation of each tissue to get the best possible overall results.

The degree and type of adaptation is going to be directly related to the mechanical, neuromuscular, and metabolic specificity. It's also important that your programs are designed to meet the specific demands of your client's daily life, health, wellness, and if applicable, athletic goals.

Now, traditionally, resistance training programs focus on developing strength in individual muscles. However, our contemporary training programs really emphasize appropriate exercise selection and progression across all muscle actions, planes of motion, and repetition tempos. It's more well-rounded in terms of an overall holistic approach to fitness in developing the body. When we talk about training adaptations in a contemporary program, they can be divided into stabilization, muscular endurance, hypertrophy, strength, and power. Let's break these down a little more.

Stabilization is the body's ability to provide optimal dynamic joint support in order to maintain correct posture, alignment, and form during all of your movements. Muscular endurance is the ability to generate and maintain force production for prolonged periods of time. Muscular hypertrophy is the enlargement of skeletal muscle fibers.

Strength is the ability of the neuromuscular system to produce internal tension. Specifically, when we talk about this kind of strength, it's in the muscles and connective tissues that pull on the bones to overcome an external force. And power is the ability of the neuromuscular system to produce the greatest possible force in the shortest possible time.

Acute variables are also important to understand. They describe the exercise training options you can use in resistance training to vary the program, such as repetitions, sets, training intensity, repetition, tempo, rest intervals, training volume, training frequency, training duration, exercise selection, and exercise order. Why are they important? Well, because acute variables determine the type, amount, and duration of stress placed on the body.

This ultimately dictates what adaptation the body will incur. So you have to be very specific and strategic to get the exact results you want, while still protecting your client's safety and minimizing risk of any kind of strain. There are many training systems you can use to structure a resistance program. You'll get a variety of effects, depending on what you choose.

Some of the most common training systems include the single set, multiple set, pyramid, superset, complex, drop set, giant set, rest-pause set, circuit training, peripheral heart action, split routine, vertical loading, and horizontal loading systems. Yep, it's a lot. The good news is, this comprehensive list gives you a lot of opportunity to introduce variety and creativity into your client programs.

One of the last things I'm going to say is something you've heard before if you've listened to other episodes. Your programming must be systematic and progressive. Learning proper exercise selection criteria and milestones for progression is critical. There's just no getting around it. It's how you keep your clients safe, engaged, and consistent for the long haul.

Specific to resistance training, NASM recommends progressing resistance training in a systematic fashion using three steps-- one, stabilization-focused exercises; two, strength-focused exercises; and three, power-focused exercises. The sequence is important, with each stage building on the one before. You must have stabilization to build strength, and you need strength to produce power. And remember, form first. The exercises should initially focus on optimizing ideal movement patterns. Once a client can demonstrate their ability to properly perform a movement, only then should you move them to the next step or level of progression.

The number one objective for fitness professionals is to commit to doing no harm to their clients. So you can never slack on things like maintaining a safe environment, proper equipment setup, using appropriate spotting techniques, and monitoring the five kinetic chain checkpoints. Aside from our intrinsic desire to provide excellent client care, you also risk hurting your own business and reputation.

On the other hand, if you dedicate yourself to becoming a master of the principles and concepts and accurately communicating and demonstrating them, you'll build lasting professional relationships with loyal clients. Even better, you'll get a front row seat to watching their lives improve. Great job so far. Keep it up. Until next time, be well, and happy studying.